Ahmed Maksud

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Summary

Wireless systems researcher with Ph.D. in Electrical Engineering (UC Riverside, 2024); expertise in **IEEE 802.11 MAC**, **physical-layer DSP and security**, and **ML/RL for wireless**, with **ns-3** simulation pipelines, WiFi-stack patches, and open-source modules. Experience with formal analysis (random-matrix theory, information theory) and implementation (C++/Python, datasheet-validated hardware models, reproducible build-to-evaluate pipelines).

Education

University of California, Riverside — Riverside, CA, USA
Ph.D. in Electrical Engineering | GPA: 4.00/4.00 — *Fall 2018 – Summer 2024*

- **Advisor:** Yingbo Hua (Fellow IEEE). **Thesis:** *Novel Methods for Wireless Network Security*

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
Bachelor of Science in Electrical and Electronic Engineering | GPA: 3.72/4.00 — *Feb 2013 – Sept 2017*

Technical Skills

Programming: Python (NumPy / SciPy, PyTorch), C / C++, MATLAB, Bash, LaTeX

Simulators & Tools: ns-3 (ns3-ai, WiFi-stack, custom modules), MATLAB toolboxes (Communications, SatComm, 5G, DSP, Deep Learning), Simulink, Cadence Virtuoso / SpectreRF

Machine Learning & RL: Deep Learning, MLP / LSTM policy networks, Autoencoder, Transformer

Wireless Standards & MAC: IEEE 802.11ax (TWT, BSR, OFDMA, action frames), CSMA/CA, channel access, QoS, action frames, MAC protocol design

PHY-Layer / DSP: OFDM, MIMO beamforming, signal estimation/detection, channel coding, S-parameters, RF impairment / calibration, multipath fading, random matrix theory, information theory

Lab Equipment: O-RAN Labkit (Firecell), Network Analyzer (Keysight E5063A), Signal Generator (Agilent N5182A), PowerCast P21XXCSR-EVB Band-6 RF rectenna

Dev Workflow: Git / GitHub, Linux, Docker / Containers, CMake, Claude Code, MCP servers, local LLMs

Experience

Texas State University — San Marcos, TX, USA
Postdoctoral Research Fellow | PI: Marcelo M. Carvalho — *Sept 2024 – present*

- **Intelligent and sustainable next-generation wireless research**: built protocol-compliant end-to-end ns-3 simulation pipelines for RL-based **Target Wake Time scheduling** and **RF energy harvesting** on **IEEE 802.11** networks with open-source **ns-3 modules** and **WiFi-stack patches**. Secured **$70K Comcast Innovation Fund** (April 2026). Hands-on **5G O-RAN xApp** experience on Firecell Labkit testbed.

University of California, Riverside — Riverside, CA, USA
Graduate Student Researcher | Advisor: Yingbo Hua — *July 2019 – Aug 2024*

- **Continuous Encryption Function (SVD-CEF) and implementations**: built, implemented, validated SVD-CEF primitives for secure **UAV-to-Ground** link via **PHY-layer encryption** and **Secret Key Generation** via **Encryption before Quantization**; achieved **20 dB BER improvement** with positive NIST randomness test.
- **MIMO Secret-Key Capacity analysis**: derived and validated closed-form SKC bounds for arbitrary MIMO channels, showing positive SKC achievable against eavesdroppers even with more antennas and better SNR than the legitimate receiver.

North-West Power Generation Company Limited — Sirajganj, Bangladesh
Assistant Engineer — *Feb 2018 | Aug 2018*

- Operations and maintenance of **225 MW thermal power plant**; capacity scheduling and load forecasting for peaking power plants.

Projects

RL driven TWT Scheduling for IEEE 802.11ax Networks [IEEE ICCCN'26]

- Developed end-to-end ns3 simulation pipeline (WiFi stack for unilateral TWT, ns3-python integration) for RL-based scheduler in **IEEE 802.11ax Target Wake Time (TWT)**. Trained **MLP-PPO** and **LSTM-PPO** agents (Stable-Baselines3) on a 208-d observation space (**BSR queue**, airtime, FCS errors, duty cycle, packets across 16 STAs) against an analytical M/D/1 baseline. −44% **energy** / +18% **throughput** / −73% **drops** (MLP-PPO, energy preset); +15% **reward** / −77% **drops** (LSTM-PPO, queue preset).
- **Tools used:** C++(ns-3,ns-3ai), Python, PPO (Stable-Baselines3).

RF Energy Harvesting & MAC simulation for IEEE 802.11 {GitHub repo}

- Designed two **ns-3** modules for sustainable IEEE 802.11: (1) datasheet-accurate **PowerCast P21XXCSR-EVB** Band 6 rectenna (−40 **to** +15 **dBm sensitivity, 0-85% efficiency curve**, 3 capacitor classes, 3 voltage classes, energy accounting, deep-discharge / overshoot protection and TX gating); (2) **IEEE 802.11 Category-127** vendor-specific action frames for AP-broadcast power-beacon schedules with autonomous STA-side parsing.
- **Tools used:** C++ (ns3), Python, IEEE 802.11 Category-127 action frames, PowerCast P21XXCSR-EVB modeling.

OFDM Optimization for Secure Communication {GitHub repo}

- Provably optimal OFDM carrier pairing (between step-1 and step-2) of a two step protocol for secure link. Power scheduling algorithms for both steps to maximize secrecy rate. **50% higher secrecy rate** with **20% lower power** seen through simulation.
- **Tools used:** Python, OFDM, joint optimization (carrier pairing, power allocation).

SVD-CEF and its application for PHY-Layer Security [IEEE SPL'22, ICC'22, SPAWC'20, Elsevier SP'23]

- Worked on new family of **physical layer cryptographic primitives** (**SVD-based Continuous Encryption Function**) and proved prior continuous-encryption candidates (DRP, IoM hashing, RP, HOP) fail security criteria.
- Designed and validated **PHY-layer DSP algorithms** for (1) secure UAV to Ground link through symbol and contellation hiding: **provable security**, **SER**$\leq$**1%** sustained under varying SNR; and (2) reliable secret key genreration through encryption before quantiation: **20 dB BER improvement**, passed **NIST randomness test**
- **Tools used:** MATLAB, Python, channel models, NIST randomness test, SVD/random-matrix/estimation theory.

Secret-Key Capacity (SKC) of MIMO Channels [IEEE WCL'24, MILCOM'23, CISS'23]

- Derived closed-form upper/lower bounds on **SKC** of MIMO channels using Maurer's bounds. Proved **positive secret-key rate is achievable even when the eavesdropper has more antennas and better SNR** than the legitimate receiver; overturning conventional wiretap-channel pessimism for MIMO physical-layer security. Introduced a Generalized Channel Probing / Pre-Processing framework for SKG over non-reciprocal channels.
- **Tools used:** Python, MATLAB, random-matrix theory, Monte-Carlo simulation, Maurer's bound analysis.

4th-Order 2.4 GHz Bandpass Filter Design & S-Parameter Analysis [Report]

- Designed and verified a **4th-order 2.4 GHz bandpass filter** in **Cadence Virtuoso (SpectreRF)** with $Q = 50$ components and $50\,\Omega$ port matching; achieved $S_{11} < -10$ **dB across passband** and Q**-factor** $= 24$, validated against 1st / 2nd-order baselines.
- **Tools used:** Cadence Virtuoso, SpectreRF, MATLAB.

Awards and Contributions

- **$70K Comcast Innovation Fund (2026)** for proposal: *Agentic TWT Scheduler for Future Home Wi-Fi*
- **Conference Travel Grant** from UCR Graduate Student Association for **IEEE MILCOM 2023**.
- **Dean's Distinguished Fellowship**, UCR Department of Electrical and Computer Engineering (2018).
- **Second Runner-up** in IEEE Video and Image Processing (VIP) Cup, 2017.
- **Honorable Mention** in International Physics Olympiad (2012).
- **Technical Reviewer** for IEEE IROS, ISIT, ICC, WCNC, ICMLCN, Globecom, WiOpt (2023–present).
- **TPC Member**, IEEE Cloud Summit 2026.